

Figure 1

# Study Analysis Pipeline

MOUSE MCAO MODEL

6 samples



10X scRNA-seq



SEURAT PROCESSING

54,599 cells, 3,000 genes → 50 PCs

SUBJECT-WISE  
TRAIN/TEST SPLIT

Train:  
sham1, sham2, mcao1, mcao2  
(37,883 cells)

Test:  
sham3, mcao3  
(16,716 cells)

8 MACHINE LEARNING MODELS

Classical ML

- Logistic Regression
- Random Forest
- XGBoost

Transformer-inspired

- scGPT
- scBERT
- scFormer
- Geneformer

Hybrid

- CNN+XGBoost

GENE IMPORTANCE EXTRACTION

Coefficients, L2 norm, SHAP → PCA loadings

MULTI-MODEL  
CONSENSUS

(≥4/6 models)

63 Consensus Genes



BIOLOGICAL VALIDATION

GO/KEGG Enrichment ❄️ ❄️

Key findings:

- IL-1 Pathway
- Dual-Interferon Signature

IL-1

IFN-α type →

IFN-γ →